

ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
 இலங்கைப் பரீட்சைத் திணைக்களம் இலங்கைப் பரීட்சைத் திணைக்களம் இலங்கைப் பரීட்சைத் திணைக்களம் இலங்கைப் பரීட்சைத் திணைக்களம் இலங்கைப் பரීட்சைத் திணைக்களம்
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අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2022(2023)
 கல்விப் பொதுத் தராதரப் பத்திர (உயர் தரப் பரீட்சை, 2022(2023)
 General Certificate of Education (Adv. Level) Examination, 2022(2023)

තොරතුරු හා සන්නිවේදන තාක්ෂණය I
 தகவல், தொடர்புபாடல் தொழினுட்பவியல் I
 Information & Communication Technology I

20 E I

පැය දෙකයි
 இரண்டு மணித்தியாலம்
 Two hours

Instructions:

- * Answer **all** the questions.
- * Write your **Index Number** in the space provided in the answer sheet.
- * Instructions are also given on the back of the answer sheet. Follow those carefully.
- * In each of the questions 1 to 50, pick one of the alternatives from (1), (2), (3), (4), (5) which is **correct or most appropriate** and mark your response on the answer sheet with a cross (X) in accordance with the instructions given on the back of the answer sheet.
- * Use of calculators is **not allowed**.

1. Which of the following statements is/are correct?

- A – Firmware is a computer program that is usually embedded in the volatile memory of a computer.
 B – A printer driver is an example for an application software.
 C – Linux is an example for a system software.

- (1) A only (2) B only (3) C only
 (4) A and B only (5) B and C only

2. Which of the following require(s) *real-time* processing?

- A – generating monthly electricity bills of customers
 B – updating the bank account balance of a customer when she/he withdraws money from an ATM
 C – updating the stock balance in a store upon successful completion of each transaction

- (1) A only (2) B only (3) C only
 (4) A and B only (5) B and C only

3. Which of the following lists a computer memory hierarchy in the **descending order** of access speed?

- (1) hard disk, registers, L2 cache, L1 cache, main memory
 (2) main memory, L1 cache, registers, L2 cache, hard disk
 (3) registers, main memory, hard disk, L1 cache, L2 cache
 (4) registers, L1 cache, L2 cache, main memory, hard disk
 (5) L1 cache, L2 cache, registers, main memory, hard disk

4. Which of the following gives the correct results of bit-wise AND and bit-wise OR operations between the two binary numbers 01010100_2 and 11101001_2 respectively?

- (1) 01000000_2 , 11111101_2
 (2) 00000010_2 , 10111001_2
 (3) 10111101_2 , 11001010_2
 (4) 11000000_2 , 00101100_2
 (5) 11111101_2 , 01010011_2

5. What is the correct binary equivalent of decimal 12.75_{10} ?

- (1) 1011.01_2 (2) 1011.11_2 (3) 1100.00_2 (4) 1100.11_2 (5) 1100.01_2

6. What is the correct 2's complement binary representation of decimal -41_{10} using 8-bits?

- (1) 00101001 (2) 01010110 (3) 10101001 (4) 11010110 (5) 11010111

7. The address of an instruction was shown as **10f9** in hexadecimal. What is that address in decimal?
 (1) 25 (2) 1249 (3) 4345 (4) 10159 (5) 16249

8. A particular command can be used to output a text file in its binary format.

Assume a file contains the following text:

0 Waste!

Referring the **Important notes** (i) and (ii) given below, select the correct output that will result when the said command is run on that file.

- (1) 00110000 00100000 01010111 01100001 01110011 01110100 01100101 00001010
 (2) 00110000 01010111 01100001 01110011 01110100 01100101 00100001 00001010
 (3) 00110000 00100000 01010111 01100001 01110011 01110100 01100101 00100001 00001010
 (4) 00110000 00100000 01110111 01100001 01110011 01110100 01100101 00100001 00001010
 (5) 00110000 00100000 01010111 01100001 01110011 01110100 01100101 00100000 00001010

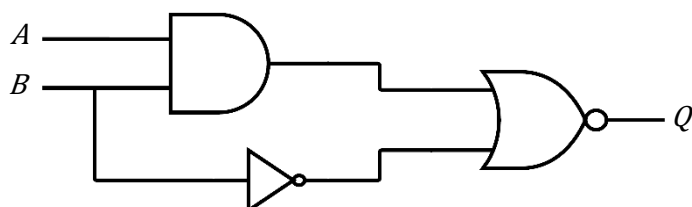
Important notes:

- (i) The file ends with a LINE FEED character.
 (ii) Some selected rows from the 7-bit ASCII table are given below:

Character	Binary
(LINE FEED)	0001010
(SPACE)	0100000
!	0100001
0	0110000
W	1010111

Character	Binary
a	1100001
e	1100101
s	1110011
t	1110100
w	1110111

9. Consider the following logic circuit:



When $B=1$, what would **definitely** be the output at Q ?

- (1) A (2) $\bar{A}$ (3) B (4) $\bar{B}$ (5) 0

10. Simplified Boolean expressions help to obtain simpler circuits.

Which of the following is a simplified form of $X + \bar{X}Y$?

- (1) X (2) Y (3) XY (4) $\bar{X}Y$ (5) $X + Y$

11. Consider the following truth table:

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

What is the correct Karnaugh map for the above truth table?

(1)

A \ BC	00	01	10	11
0	0	0	1	1
1	1	1	0	0

(2)

A \ BC	00	01	11	10
0	0	0	1	1
1	1	1	0	0

(3)

A \ BC	00	10	01	11
0	0	1	0	1
1	1	0	1	0

(4)

A \ BC	00	10	11	01
0	0	1	1	0
1	1	0	0	1

(5)

A \ BC	00	11	10	01
0	0	1	1	0
1	1	0	0	1

12. A program in execution in a computer is called a process. Such a process transits between several states during its lifetime. Which of the following correctly represents a possible state transition sequence of a process?

- (1) New $\rightarrow$ Ready $\rightarrow$ Running $\rightarrow$ Waiting $\rightarrow$ Ready $\rightarrow$ Running $\rightarrow$ Terminated
- (2) New $\rightarrow$ Ready $\rightarrow$ Waiting $\rightarrow$ Running $\rightarrow$ Waiting $\rightarrow$ Running $\rightarrow$ Terminated
- (3) New $\rightarrow$ Running $\rightarrow$ Ready $\rightarrow$ Waiting $\rightarrow$ Running $\rightarrow$ Ready $\rightarrow$ Terminated
- (4) New $\rightarrow$ Running $\rightarrow$ Waiting $\rightarrow$ Ready $\rightarrow$ Waiting $\rightarrow$ Running $\rightarrow$ Terminated
- (5) New $\rightarrow$ Waiting $\rightarrow$ Running $\rightarrow$ Ready $\rightarrow$ Running $\rightarrow$ Ready $\rightarrow$ Terminated

13. Which of the following is **not** a task of a computer operating system?

- (1) selecting a memory *frame* for a page of a process
- (2) maintaining a list of free memory *frames*
- (3) maintaining a *page table* for each process
- (4) monitoring the usage of binary files on a hard disk
- (5) swapping processes between main memory and hard disk

14. The block size of a disk is 4KB. A portion of its File Allocation Table (FAT) at a particular time is shown below. The portion shown indicates the blocks of the *average.py* file as well.

FAT

200	202
201	200
202	-1
203	201
204	205

- Notes:** I. The last block of a file is indicated by -1.
II. The *directory entry* of a file contains the block number of the first block of the file.

Which of the following gives the *directory entry* for the *average.py* file **and** the disk space allocated for the *average.py* file respectively?

- (1) 200, 12KB (2) 200, 16KB (3) 200, 20KB (4) 203, 16KB (5) 203, 20KB

15. Which of the following are *Transport Layer* protocols of the TCP/IP stack?

- A – Transmission Control Protocol (TCP)
B – User Datagram Protocol (UDP)
C – File Transfer Protocol (FTP)
D – Internet Protocol (IP)

- (1) A and B only (2) A and C only (3) B and C only
(4) B and D only (5) All A, B, C and D

16. Which of the following statements is/are correct about MAC and IPv4 addresses?

- A – MAC addresses are 32 bits in length and are used in the network layer.
B – MAC addresses are 48 bits in length and are used in the datalink layer.
C – IPv4 addresses are 32 bits in length and are used in the network layer.

- (1) A only (2) B only (3) C only
(4) A and C only (5) B and C only

17. Which of the following is/are correct regarding a *firewall*?

- A – It can monitor and filter outgoing traffic from an internal network.
B – It protects a network from unauthorized accesses.
C – It can be a hardware, a software or a combination of both.

- (1) A only (2) A and B only (3) A and C only
(4) B and C only (5) All A, B and C

18. An organization with the assigned IP address block 193.1.1.0/24 needs to define **eight** subnets. Each subnet should provide for more than 25 IP addresses. Which of the following correctly lists the number of bits needed to identify the given network, the total number of bits needed to identify the subnets, and the number of bits needed to assign unique IP addresses for this requirement, respectively?

- (1) 24, 3,5 (2) 24, 5, 3 (3) 24, 27, 5 (4) 27, 3,5 (5) 27, 30, 2

19. Which of the following statements is correct regarding network topologies?

- (1) In *bus topology*, a central network hub is used to connect all nodes.
(2) In *star topology*, a linear cable is used to connect all nodes.
(3) In *ring topology*, messages are sent only clockwise.
(4) In *ring topology*, each node is directly connected only to two of its neighbors.
(5) In *mesh topology*, each node is always connected to one other node only.

20. Consider the seven layer OSI reference model and match each of the given layers labeled from P to S to the corresponding responsibility of it labeled from 1 to 4.

Layer
P – Application layer
Q – Physical layer
R – Transport layer
S – Network layer

Responsibility
1 – binary transmission over the communication medium
2 – route determination
3 – user services that include file transfer, remote access etc.
4 – data delivery from process to process

- (1) P – 1, Q – 3, R – 2, S – 4
 (2) P – 2, Q – 4, R – 3, S – 1
 (3) P – 3, Q – 1, R – 2, S – 4
 (4) P – 3, Q – 1, R – 4, S – 2
 (5) P – 4, Q – 2, R – 1, S – 3

21. Which of the following statements is/are correct?

A – A *digital signature* ensures the authenticity of a message.

B – In *asymmetric key encryption*, different keys are used for encryption and decryption.

C – The encryption process transforms plaintext to ciphertext.

- (1) A only
 (2) B only
 (3) C only
 (4) A and B only
 (5) All A, B and C

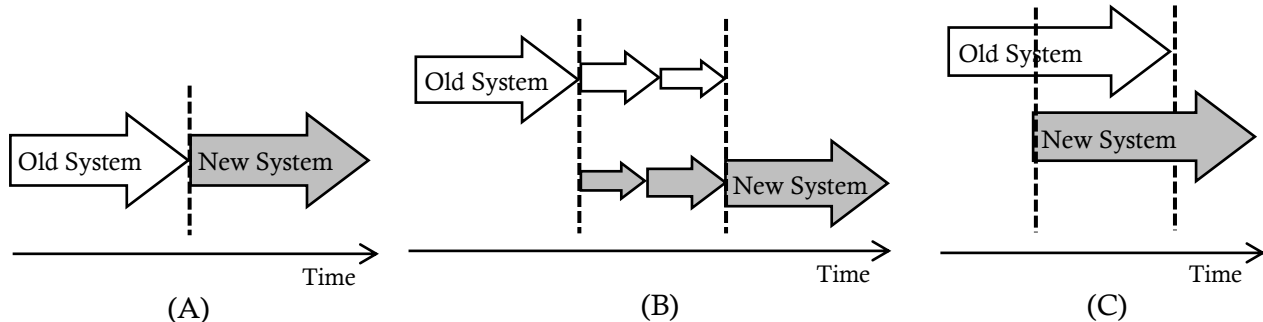
22. Match each of the given data communication protocols labelled from P to T to the corresponding descriptions labelled from 1 to 5.

Protocol
P – Hyper Text Transfer Protocol (HTTP)
Q – Transmission Control Protocol (TCP)
R – Domain Name System (DNS) Protocol
S – Internet Protocol (IP)
T – User Datagram Protocol (UDP)

Description
1 – provides directory lookup service for given web addresses and URLs
2 – provides a very reliable data transfer service
3 – used in the world wide web
4 – provides a connection-less transport service
5 – handles unique addressing of hosts in the Internet

- (1) P – 2, Q – 4, R – 1, S – 5, T – 3
 (2) P – 2, Q – 5, R – 4, S – 1, T – 3
 (3) P – 3, Q – 2, R – 1, S – 5, T – 4
 (4) P – 3, Q – 4, R – 5, S – 1, T – 2
 (5) P – 4, Q – 2, R – 3, S – 1, T – 5

23. The following diagrams labelled (A), (B) and (C) illustrate three software deployment types.



Which of the following correctly represents (A), (B) and (C) deployment types respectively?

- (1) Direct, Phased and Parallel
 (2) Direct, Pilot and Parallel
 (3) Parallel, Phased and Direct
 (4) Parallel, Pilot and Phased
 (5) Phased, Direct and Pilot

24. Consider the information system types in **List A** and the descriptive examples in **List B**. Identify the most suitable matching between the items in lists **A** and **B**.

List A
A1 - Decision Support System (DSS)
A2 - Content Management System (CMS)
A3 - Transaction Processing System (TPS)

List B
B1 - a system that allows to update, create, and manage the details in a news website
B2 - a system that handles electronic fund transfers
B3 - a system that combines data and analytical tools for sales forecasting based on historical data

- (1) A1 – B1, A2 – B2, A3 – B3
 (2) A1 – B2, A2 – B1, A3 – B3
 (3) A1 – B2, A2 – B3, A3 – B1
 (4) A1 – B3, A2 – B1, A3 – B2
 (5) A1 – B3, A2 – B2, A3 – B1

25. Which of the following statements is/are correct regarding System Development Life Cycle (SDLC) models?

A – In the *agile* model, small portions of systematically developed working software are delivered to the client frequently.

B – Late changes in the requirements can be easily accommodated in the *waterfall* model.

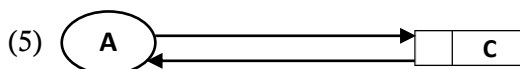
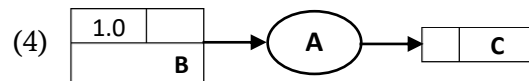
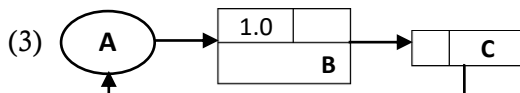
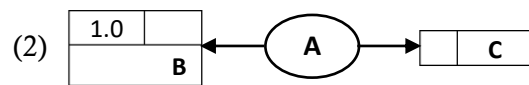
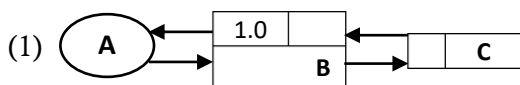
C – *Prototyping* model can be practiced without client interactions.

- (1) A only
 (2) B only
 (3) C only
 (4) A and B only
 (5) A and C only

26. *Non-functional requirements* specify quality attributes of a system. Which of the following is an example for a *non-functional requirement*?

- (1) the email system should allow users to attach files
 (2) each page of the website must load within 4 seconds
 (3) administrator of the E-commerce website should be able to view a list of customers
 (4) a user of the online banking system should be able to view the last transactions
 (5) the ATM machine should allow users to print a receipt

27. Which of the following Data Flow Diagrams (DFDs) is correct with respect to the rules of data flow modelling? (Note: A – an external entity, B – a process, C – a data store)



28. Which of the following statements is correct regarding *software testing*?

- (1) Integration testing is usually carried out before unit testing.
 (2) Black-box testing techniques are usually used in acceptance testing.
 (3) White box testing examines the behaviour of a software based only on the inputs to a system.
 (4) Unit testing examines the entire system's functionality as a whole.
 (5) System testing is usually carried out after the user acceptance testing.

29. Consider the following relational schema:

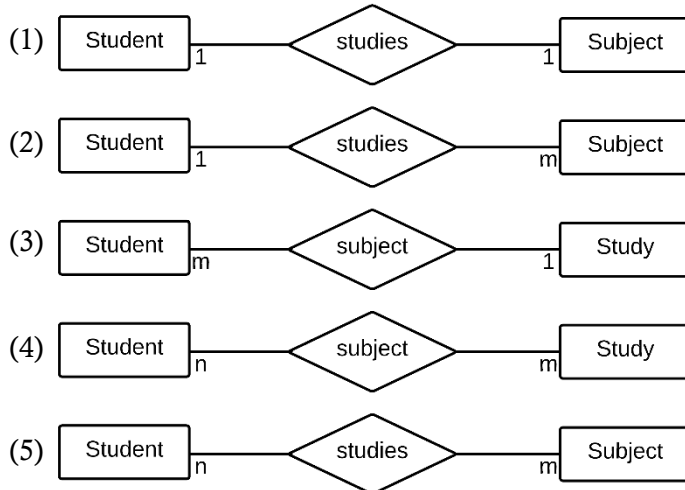
Student (StudentId, StudentName, Address, Gender, DateOfBirth)

Study (StudentId, SubjectId, Grade)

Subject (SubjectId, SubjectName)

Which of the following is the most suitable *Entity Relationship (ER)* diagram to correctly represent the relationship between **Student** and **Subject** entities?

Note: In the ER diagrams, the entities are drawn without attributes.



- The tables which are partially extracted from a database used in an information system developed for a shop are shown below. Answer the questions from 30 to 32 using those tables.

Customer

CusId	Fname	Lname	Location
C001	Saman	Perera	Dehiwala
C002	Kalum	Gamage	Galle
C003	Shiromi	Silva	Galle
C004	Kalum	Perera	Kandy

Product

ProId	Name
PR001	Refrigerator
PB401	Blender
PM025	Mobile Phone
PP009	Inkjet Printer

Order

OrderId	CusId	OrderDate	SellerId
A001	C002	2022-07-14	S001
A002	C003	2022-07-14	S001
A003	C002	2022-07-18	S002
A004	C004	2022-07-20	S002

Order_Product

OrderId	ProId
A003	PR001
A001	PR001
A002	PB401
A003	PM025
A004	PP009

30. Which of the following shows the most suitable *primary keys* for **Order** and **Order_Product** relations?

- | | |
|-------------------------------------|--|
| (1) Order: CusId, | Order_Product: OrderId |
| (2) Order: OrderId, | Order_Product: OrderId |
| (3) Order: OrderId, | Order_Product: OrderId + ProdId |
| (4) Order: CusId + SellerId, | Order_Product: ProdId |
| (5) Order: OrderId + CusId, | Order_Product: OrderId |

31. What will be the output after executing the following SQL statement?

```
SELECT Customer.Fname, Customer.Lname, Order.OrderId
FROM Customer INNER JOIN Order ON Customer.CusId = Order.CusId
WHERE Customer.Location="Galle";
```

(1)

Fname	Lname	OrderId
Kalum	Gamage	A001
Kalum	Gamage	A003
Shiromi	Silva	A002

(2)

Fname	Lname	OrderId
Kalum	Gamage	A004
Kalum	Perea	A001
Kalum	Gamage	A003
Shiromi	Silva	A002

(3)

Fname	Lname	OrderId
Kalum	Gamage	A001
Kalum	Perera	A003
Shiromi	Silva	A002

(4)

Lname	Fname	OrderId
Gamage	Kalum	A001
Gamage	Kalum	A003
Silva	Shiromi	A002

(5)

Fname	Lname	OrderId
Kalum	Gamage	A001
Shiromi	Silva	A002

32. Which of the following is correct regarding the **Order** relation?

- (1) CusId attribute uniquely identifies each tuple in the relation.
- (2) The relation is in First Normal Form (1NF).
- (3) The relation is in Second Normal Form (2NF).
- (4) Orders of each customer are handled by a unique salesperson.
- (5) The relation consists of a composite primary key.

33. Which of the following statements is/are correct regarding the *normalization* concept?

- A – In first normal form, atomic attributes are **removed** from a relation.
 B – In second normal form, partial dependency of attributes on the primary key are **removed**.
 C – In third normal form, transitive dependency of attributes are **removed**.

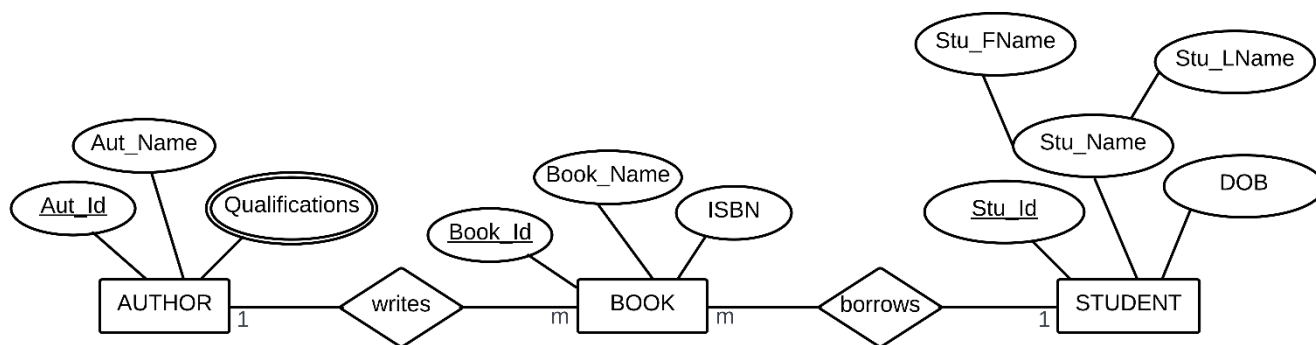
- (1) B only
- (2) A and B only
- (3) A and C only
- (4) B and C only
- (5) All A, B and C

34. Which of the following statements is/are correct regarding *Entity Relationship (ER) modelling*?

- A – A weak entity is dependent on another entity.
 B – A derived attribute is represented as an attribute in a relation.
 C – An entity can contain a multi-value attribute and a composite attribute at the same time.

- (1) A only
- (2) B only
- (3) A and C only
- (4) B and C only
- (5) All A, B and C

35. The following ER diagram represents a scenario of students borrowing books from a library. Which of the following gives the most suitable relation list for the given ER diagram?

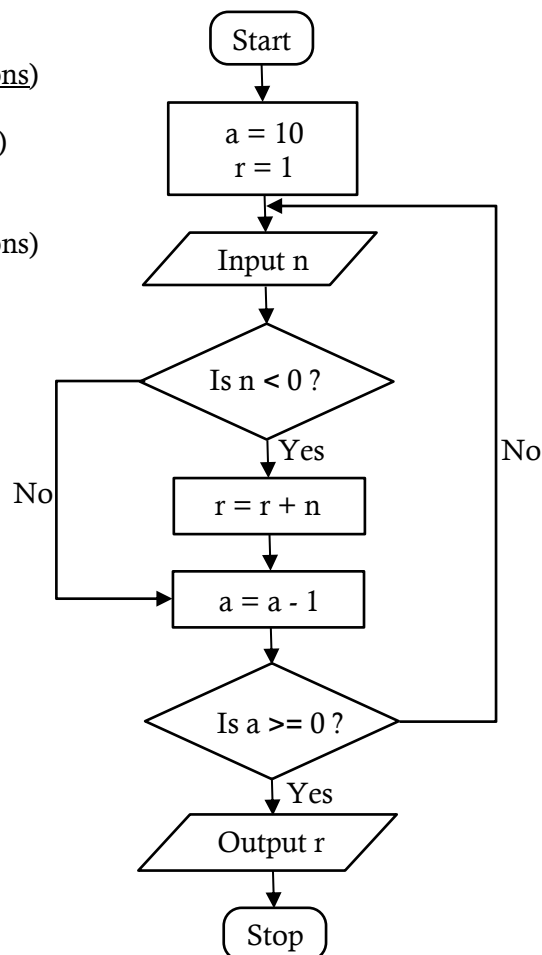


- (1) BOOK (Book_Id, Book Name, ISBN, Stu_Id, Aut_Id)
STUDENT (Stu_Id, Stu_FName, Stu_LName, DOB)
AUTHOR (Aut_Id, Aut_Name)
AUTHOR_QUALIFICATION (Aut_Id, Qualifications)
- (2) BOOK (Book_Id, Book_Name, ISBN)
STUDENT (Stu_Id, Stu_LFName, Stu_LName, DOB)
AUTHOR (Aut_Id, Aut_Name)
AUTHOR_QUALIFICATION (Aut_Id, Qualifications)
- (3) BOOK (Book_Id, Book Name, ISBN, Stu_Id, Aut_Id)
STUDENT (Stu_Id, Stu_FName, Stu_LName, DOB)
AUTHOR (Aut_Id, Aut_Name, Qualifications)
- (4) BOOK (Book_Id, Book Name, ISBN, Stu_Id, Aut_Id)
STUDENT (Stu_Id, Stu_Name, DOB)
AUTHOR (Aut_Id, Aut_Name)
AUTHOR_QUALIFICATION (Aut_Id, Qualifications)
- (5) BOOK (Book_Id, Book Name, ISBN, Stu_Id, Aut_Id)
STUDENT (Stu_Id, Stu_Name, DOB)
AUTHOR (Aut_Id, Aut_Name)
AUTHOR_QUALIFICATION (Aut_Id, Qualifications)
BORROW (Aut_Id, Book_Id)
WRITE (Aut_Id, Book_Id)

36. Which of the following statements is/are correct about the algorithm expressed by the given flowchart?

- A – An input is taken from the user only once.
B – The output of the algorithm is always 9.
C – The algorithm outputs the summation of all the numbers entered.

- (1) A only
- (2) B only
- (3) C only
- (4) A and B only
- (5) B and C only



37. What would be the output of the following Python code if the input was 25?

```
x = int(input())
x = (x % (x - 21)) **3
print (x)
```

- (1) 0 (2) 1 (3) 3 (4) 12 (5) 25

38. What would be the output of the following Python code?

```
def fun(para1, para2):
    x=foo(para2, para1)
    return x

def foo(para3, para4):
    return para3 - para4

result=fun(2, 4)
print("Result is " + str(result))
```

- (1) Result is 0 (2) Result is 2 (3) Result is -2
(4) Result is (2, 4) (5) Result is +2

39. What would be the output of the following Python code?

```
def foo(name, age=18, address="Kandy"):
    print (name, address, age)

foo("Nimal", 25, "Colombo")
```

- (1) Nimal Colombo 25 (2) Nimal, Colombo, 25
(3) Nimal, Kandy, 18 (4) Nimal Kandy 18
(5) Nimal 18 Kandy

40. What would be the output of the following Python code?

```
numbers=[10, 20, 30, 40, 50]
numbers.pop(1)
numbers.append(60)
numbers.pop(2)
print(numbers)
```

- (1) [10, 50, 60] (2) [10, 20, 40, 60] (3) [10, 30, 50, 60]
(4) [20, 30, 40, 50] (5) [20, 30, 50, 60]

41. What would be the output of the following Python code?

```
val = 9
for i in range(5):
    for j in range(2, 3, 1):
        val += 1
        if (val % 2) == 0:
            continue
        val += 2
    else:
        val += 2
print(val)
```

- (1) 18 (2) 24 (3) 29 (4) 38 (5) 39

42. Which of the following is/are correct regarding Python functions?

- A – A Python function can return a data structure that contains multiple values.
 B – A Python function can be used without passing any parameters to it.
 C – Parameters can be passed to a python function by value or by reference.

- (1) B only (2) C only (3) A and C only
 (4) B and C only (5) All A, B and C

43. Which of the following HTML tags can be used to change the appearance of a word in a text?

- (1) <i>, , ,
 (2) , <i>, , <h1>
 (3) , , <sup>, (4) <i>, <u>,
, <sup>
 (5) <u>, <i>, ,

44. What would be the output of the following HTML code segment?

```
<dl>
  <dt> Vegetable </dt>
  <dd> Potato </dd>
  <dt> Fruit </dt>
  <dd> Orange </dd>
</dl>
```

- (1) • Vegetable
 • Potato
 • Fruit
 • Orange
- (2) Vegetable
 Potato
 Fruit
 Orange
- (3) • Vegetable
 Potato
 • Fruit
 Orange
- (4) 1. Vegetable
 Potato
 2. Fruit
 Orange
- (5) • Vegetable
 – Potato
 • Fruit
 – Orange

45. Which of the following statements is/are correct regarding HTML and CSS?

- A – CSS can be used to describe how HTML elements are to be displayed on screen.
 B – External CSS can be used to define the style for many HTML pages.
 C – Inline CSS can be used to apply a style to a single HTML element.

- (1) A only (2) A and B only (3) A and C only
 (4) B and C only (5) All A, B and C

46. Which of the following HTML code line can be used to create a hyperlink to the website of the National Institute of Education? (The URL of the website is <http://nie.lk>)

- (1) <a src = <http://nie.lk>>National Institute of Education
 (2) <a href = "<http://nie.lk>">National Institute of Education
 (3) <a img = <http://nie.lk>>National Institute of Education
 (4) <a href = "<http://nie.lk>"National Institute of Education>
 (5) <a src= <http://nie.lk>National Institute of Education>

47. Which of the following could be used to create an array in PHP?

- A – \$city[] = array("Colombo");
 B – city[] = "Colombo";
 C – \$city = array("Colombo");

- (1) A only (2) B only (3) C only
 (4) A and C only (5) B and C only